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PROJECT NAME: SMART ASSIGNMENT SUBMISSION (SAS)

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INTRODUCTION

The rapid growth of digital learning environments has increased the need for efficient systems that allow students to submit assignments electronically. Traditional methods such as email submissions or physical documents often result in problems such as lost files, missed deadlines, and lack of centralized management. Universities require systems that simplify academic workflows while ensuring security and accessibility.

The Smart Assignment Submission (SAS) System is a mobile-friendly web application designed to facilitate assignment submission and tracking for students. The system allows students to upload assignments, view deadlines, and monitor their academic progress using a simple mobile interface.

The application is optimized for mobile devices, allowing students to access the system from smartphones without installing additional software. This approach increases accessibility and ensures that the platform can be used in environments where computers may not always be available.

1.1 Purpose

The purpose of the SAS system is to create a centralized platform that improves assignment management within the university environment. The system aims to enhance communication between students and instructors while providing a reliable platform for academic document submission.

Objectives of the system include:

- Allow students to create accounts and log in securely
- Enable assignment upload and management
- Allow students to track deadlines and grades
- Provide a responsive mobile interface

2 Requirements

2.1 Functional Requirements

No	Requirement	Description
1	User Registration	Students can create accounts using their registration numbers
2	User Login	Students log into the system using credentials
3	Assignment Submission	Students upload assignments to the system
4	Assignment Management	Students can edit or delete submissions
5	View Grades	Students can see grades assigned by instructors
6	Deadline Tracking	Students can monitor upcoming deadlines
7	Admin Management	Administrator manages users and assignments

Table 1 Functional Requirements

2.2 Non-Functional Requirements

No	Requirement	Description
1	Mobile Compatibility	System optimized for smartphones
2	Security	Passwords must be securely stored
3	Performance	System responds quickly to user requests
4	Usability	Interface must be simple and easy to use
5	Reliability	System should store data without loss

Table 2 Non-Functional Requirements

3 Design and Implementation

3.1 Overview

The Smart Assignment Submission system is implemented as a mobile-friendly web application using standard web technologies. The frontend of the system is developed using HTML5, CSS3, and JavaScript. The application is designed to run directly within a web browser without requiring installation.

The system uses localStorage to manage user sessions and store assignment information. This approach allows the application to function without a dedicated backend server while still providing persistent data storage for users.

3.2 Use Case Diagram

The use case diagram describes interactions between the system actors and the SAS application. The primary actors include Guest users, Registered Students, and Administrators.

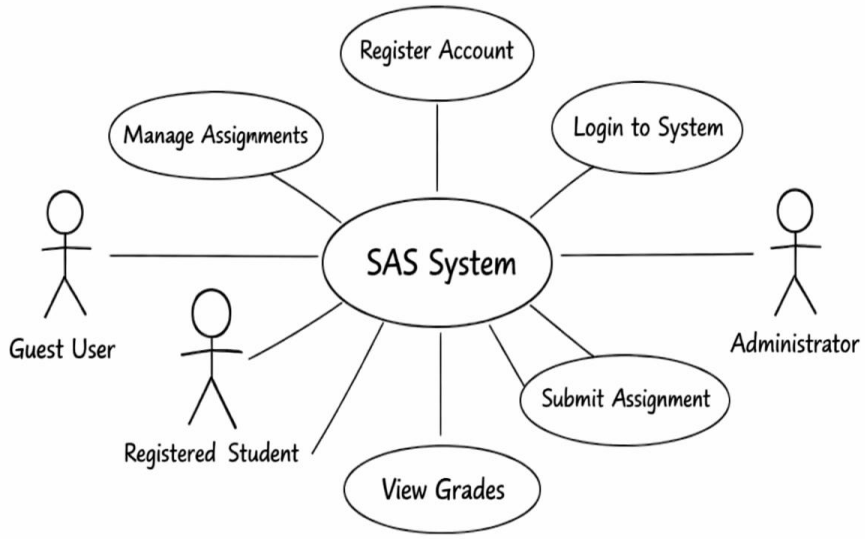

Figure 1 Use Case Diagram

3.3 Sequence Diagram

The sequence diagram illustrates the process that occurs when a student submits an assignment. The student logs into the system, selects an assignment file, uploads it to the system, and the system validates and stores the submission.

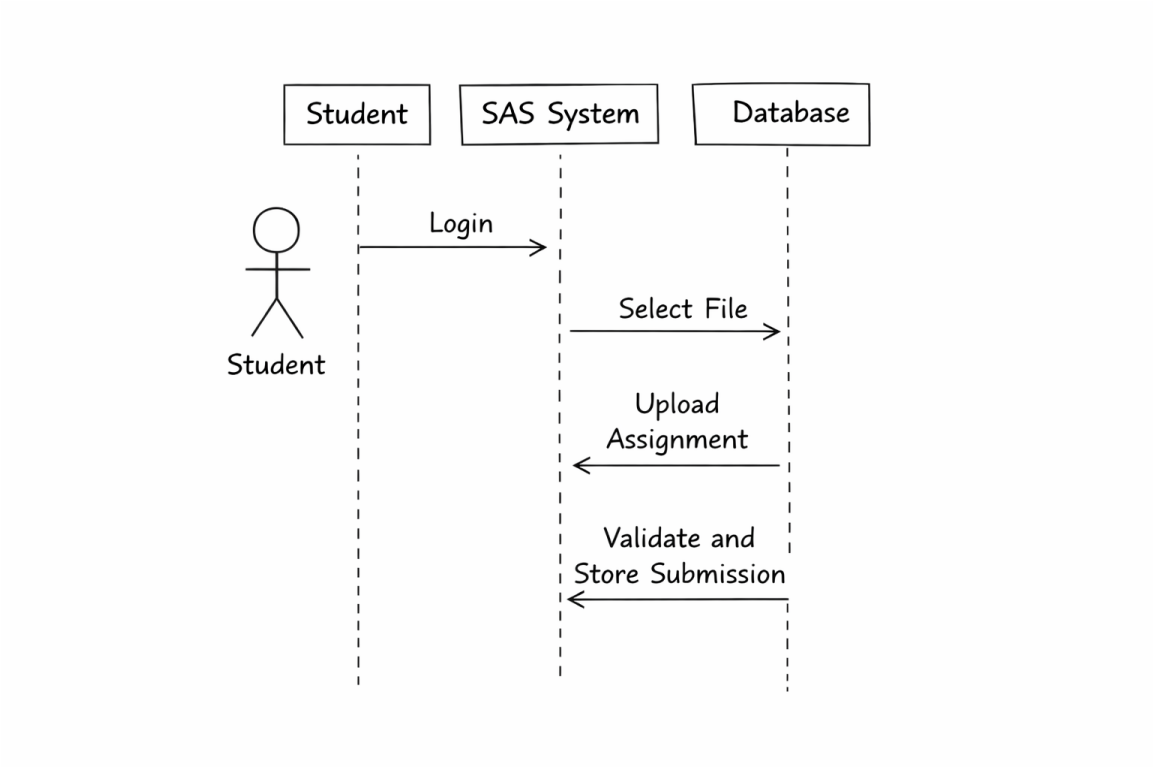

Figure 2 Sequence Diagram

3.4 Database Schema

The SAS system stores data using localStorage. The stored data includes information related to users, assignments, grades, and deadlines.

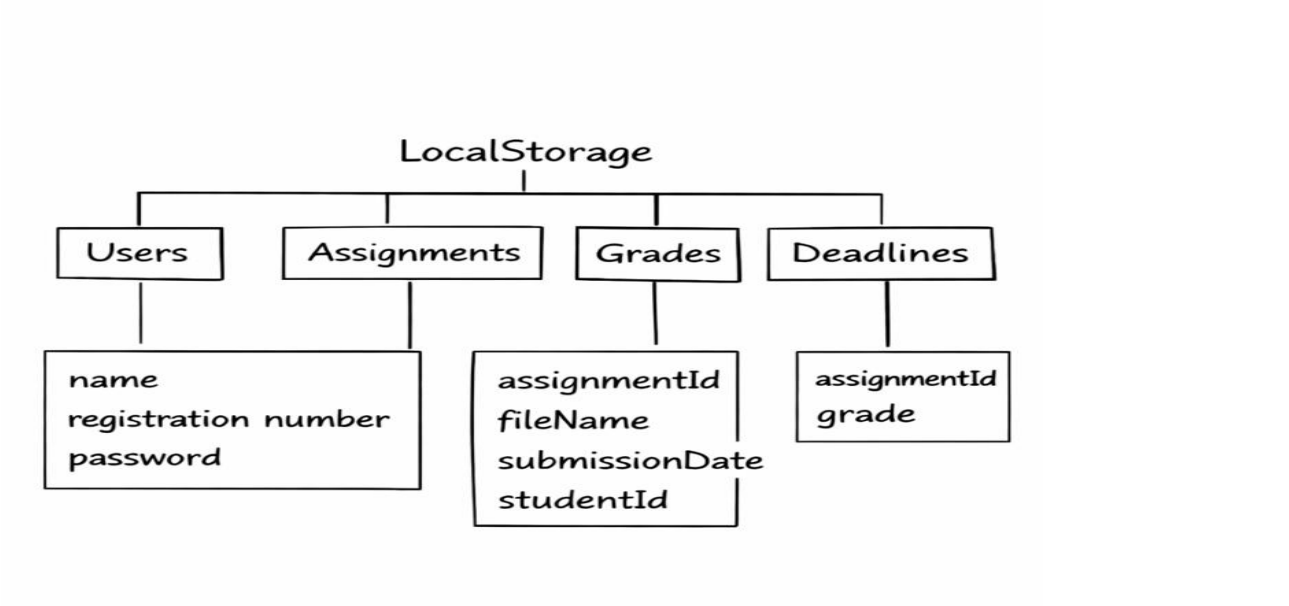

Figure 3 Database Schema

3.5 API Description

Method	Endpoint	Access	Description
POST	/api/auth/register	Public	Register new account
POST	/api/auth/login	Public	Login to system
GET	/api/assignments	User	Retrieve assignments
POST	/api/assignments	User	Upload assignment
PUT	/api/assignments/:id	User	Update assignment
DELETE	/api/assignments/:id	User	Delete assignment

Table 3 API Description

3.6 User Authentication

The system implements authentication using session-based logic. When a user logs in, a session identifier is generated using the registration number and a timestamp. This session is stored in localStorage to maintain login persistence.

Security measures include password verification during login and automatic redirection of unauthorized users attempting to access protected pages.

3.7 User Authentication

Authentication is a crucial component of the Smart Assignment Submission (SAS) system. It ensures that only authorized users can access sensitive data and system features.

3.8 JWT Overview

JSON Web Tokens (JWT) is a widely adopted method for securely transmitting information between parties as a JSON object. In the SAS system, JWTs are used to verify the identity of users during login and to protect access to restricted resources.

When a user logs in successfully, a JWT is generated and returned to the client. This token is then used for authenticating subsequent requests, ensuring that only authorized users can access protected pages such as the dashboard or submission page.

3.8.1 Login Process

1. **User Registration:** Students register by providing their full name, registration number, and password.
2. **Token Generation:** After successful registration or login, a unique JWT is created for the user. This token contains encoded information about the user's identity and session.
3. **Token Storage:** The JWT is securely stored on the client side, using localStorage in the browser.
4. **Authentication:** To access protected routes, the user must include the JWT with each request. The server verifies the token's signature and grants access only if the token is valid.

3.8.2 Security Measures

- **Password Hashing:** Passwords are hashed before storage to prevent plain-text exposure.
- **Token Expiration:** JWTs are issued with a limited lifetime to reduce risk of misuse.
- **Protected Routes:** Server-side checks ensure that only requests with valid tokens can access sensitive endpoints.

4 Results

The SAS system has been successfully implemented according to the specified requirements. The application provides a responsive mobile interface that allows students to submit assignments and monitor their academic progress.

Main system pages include:

• Login Page

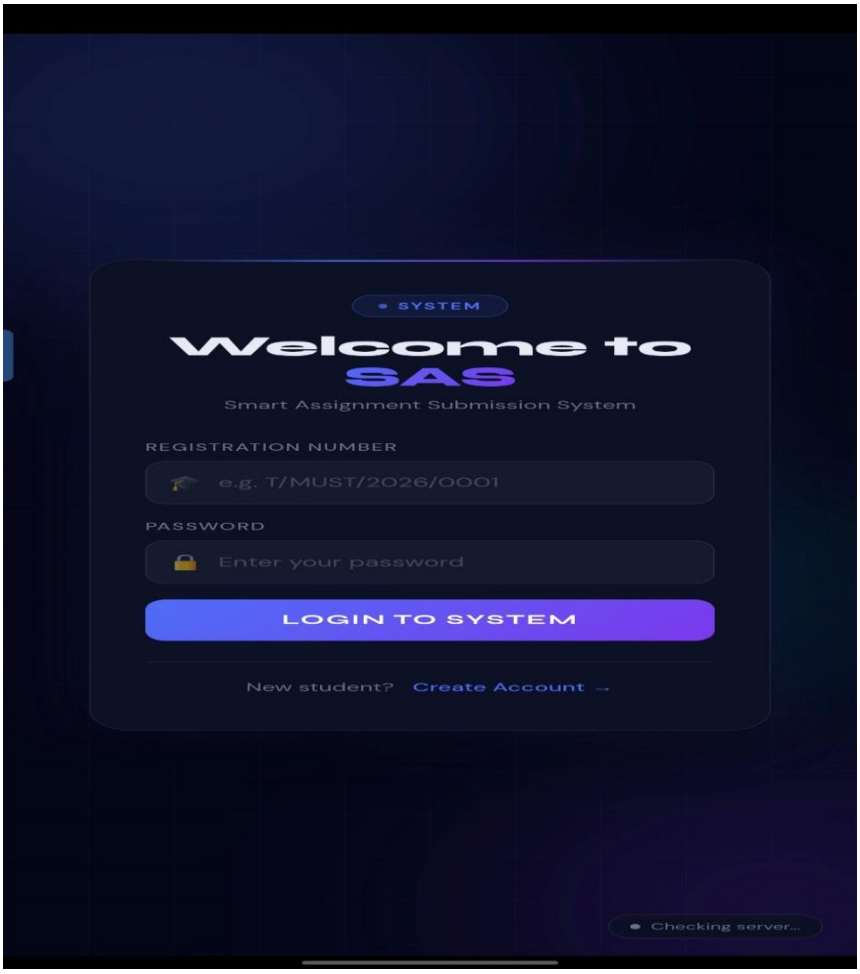

Figure 4 Login Page

Registration Page

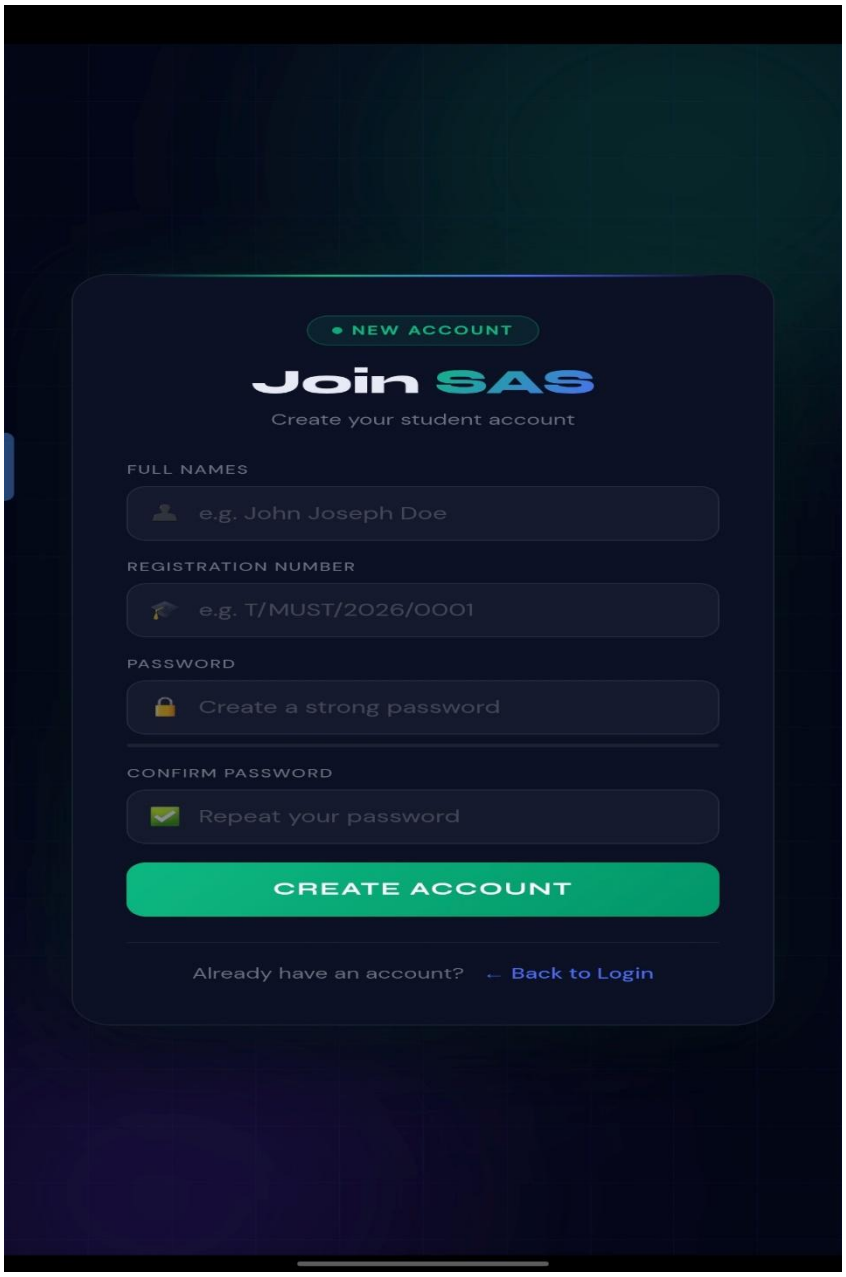

Figure 5 Registration Page

• Main Dashboard

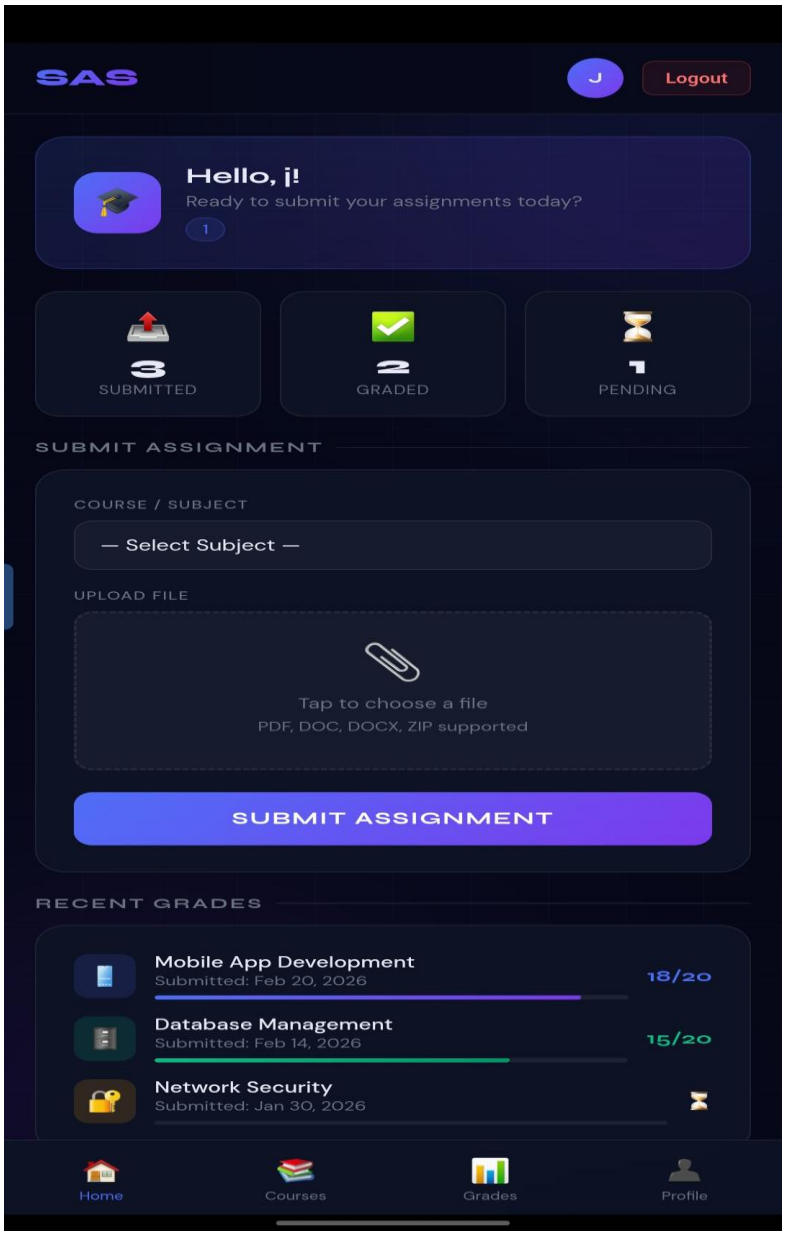

Figure 6 Main Dashboard

• Assignment Submission Page

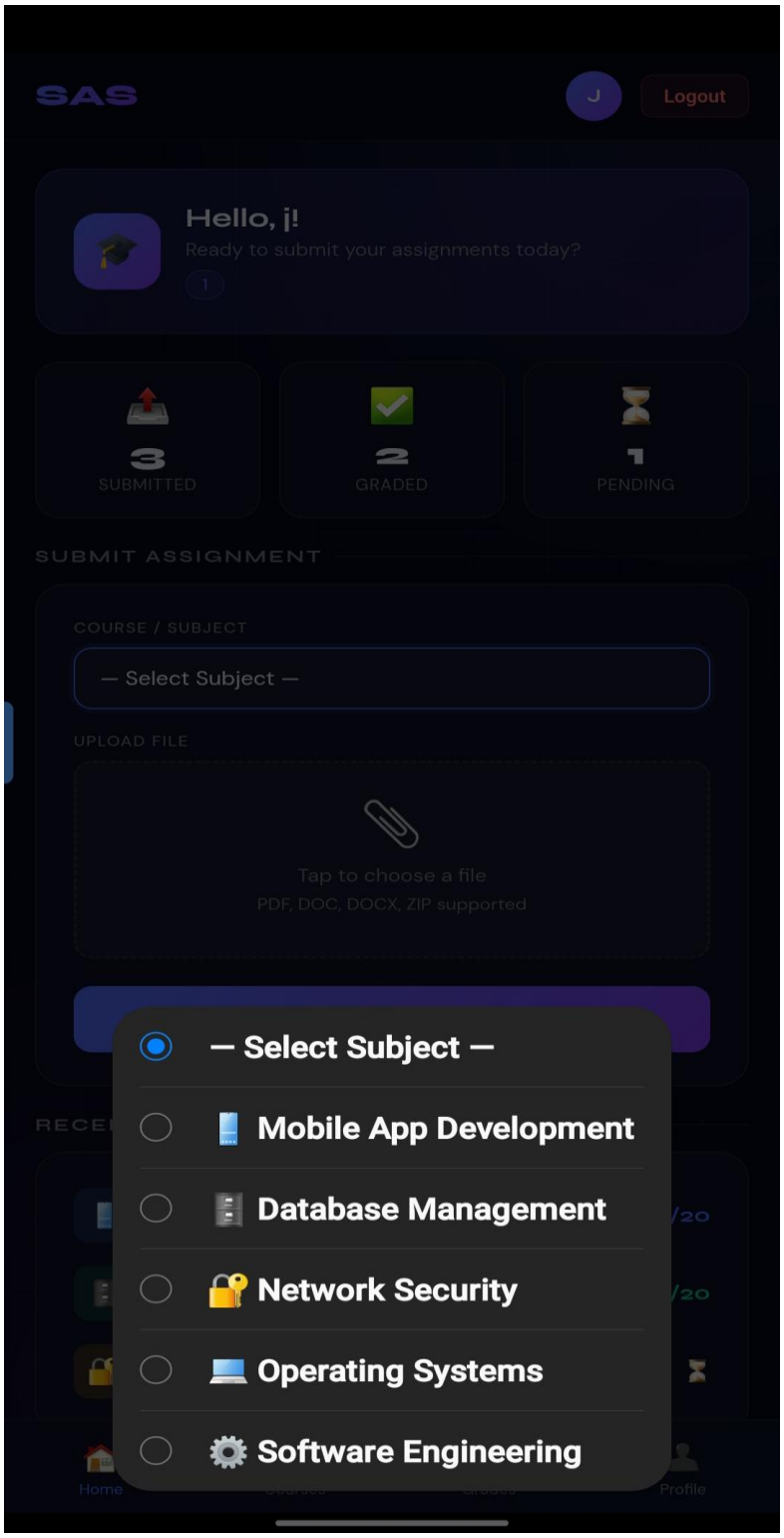

Figure 7 Assignment Submission Page

• Grades and Deadlines Page

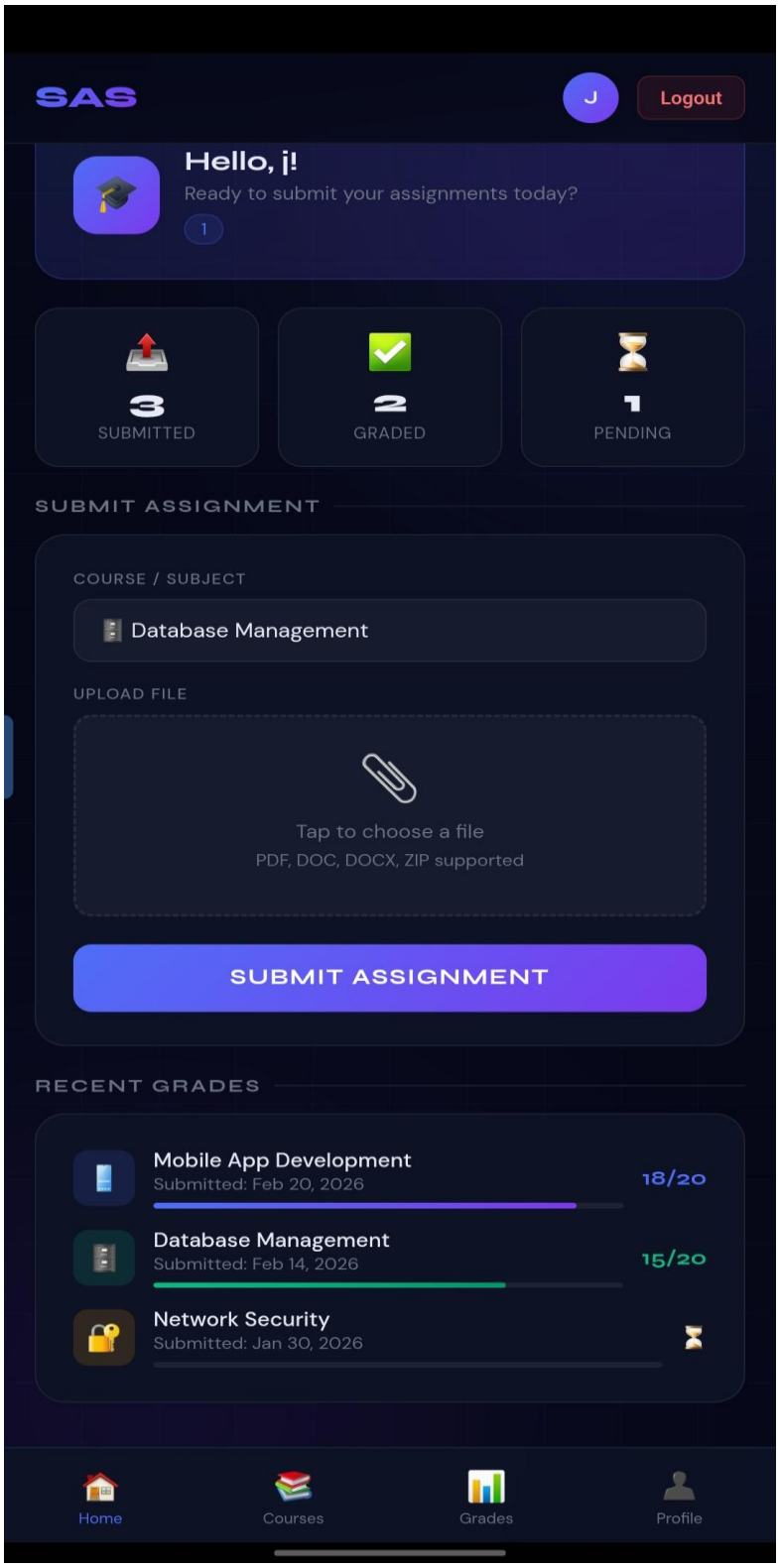

Figure 8 Grades and Deadlines Page

SUMMARY AND CONCLUSION

The Smart Assignment Submission system provides an effective solution for managing assignment submissions within a university environment. The system simplifies the process of uploading assignments and tracking academic progress.

Advantages of the system include improved organization of academic work, easier communication between students and instructors, and mobile accessibility.

Future improvements may include integration with a backend server, implementation of a cloud database, and additional features such as instructor grading tools.

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